Software Configuration Management(SCM)
Chapter 7

- Concept
- Requirement and element of SCM
- Baseline
- SCM Repository
- Versioning and version control
- SCM Process
- Change control process

Software Configuration Management(SCM)

- The art of identifying, organizing, and controlling modifications to the software being built.
- The main goal is to minimize mistakes.
- It is a umbrella activity, applied throughout the software engineering process.
- Auditing of the software configuration
- Reporting of all the changes applied to configuration
- Control of change

Requirement and element of SCM

- Identification, control, audit, and status accounting are the four basic requirements for a software configuration management system.
- **Identification:**
  - Each software part is labeled so that it can be identified.
  - Furthermore, there will be different versions of the software parts as they evolve over time, so a version or revision number will be associated with the part.
  - The key is to be able to identify any and all artifacts that compose a released configuration item.
  - It helps to answer the following question.
    - What are the component of the product?
    - What are the version of configured item?
  - Think of this as a bill of materials for all the components in your automobile. When the manufacturer realizes that there has been a problem with parking brakes purchased from a subcontractor, it needs to know all the automobile models using that version of the parking brake.

Control

- In the context of configuration management, “control” means that proposed changes to a CI (Configuration Item) are reviewed and, if approved, incorporated into the software configuration.
- The goal is to make informed decisions and to acknowledge the repercussions associated with a change to the system.
- These changes may impact budgets, schedules, and associated changes to other components
- It answer the question:
  - Which item are controlled?
  - How are changes controlled?
  - Who controls the change?

Audit

- Auditing an SCM system means that approved requested changes have indeed been implemented.
- The audits allow managers to determine whether software evolution is proceeding both logically and in conformance with requirements for the software.
- The SCM system should document changes, versions, and release information for all components of each configuration item.
- When such documentation is in place, auditing becomes a straightforward analysis task.
- It verify the correctness of the product.

Status Accounting

- This accounting provides the historic information to determine both what happened and when on the software project.
- Status accounting enables the auditing requirement of the SCM.
- As a project manager, the status accounting holds a wealth of information on the amount of effort required throughout the life cycle of the product in its development and maintenance.
- This is critical to the software project manager in making estimates for new systems based on historic information.
- The SCM can be used as one of the key components of the project managers' metrics system.

Baseline

- A baseline is a formally accepted version of a software configuration item. It is designated and fixed at a specific time while conducting the SCM process. It can only be changed through formal change control procedures.
- **Activities during this process:**
  - Facilitate construction of various versions of an application
  - Defining and determining mechanisms for managing various versions of these work products
  - The functional baseline corresponds to the reviewed system requirements
  - Widely used baselines include functional, developmental, and product baselines
- In simple words, baseline means ready for release.

Baseline

- A baseline is a milestone in the development of software that marked the delivery of one or more software configuration items.
- IEEE define baseline as:

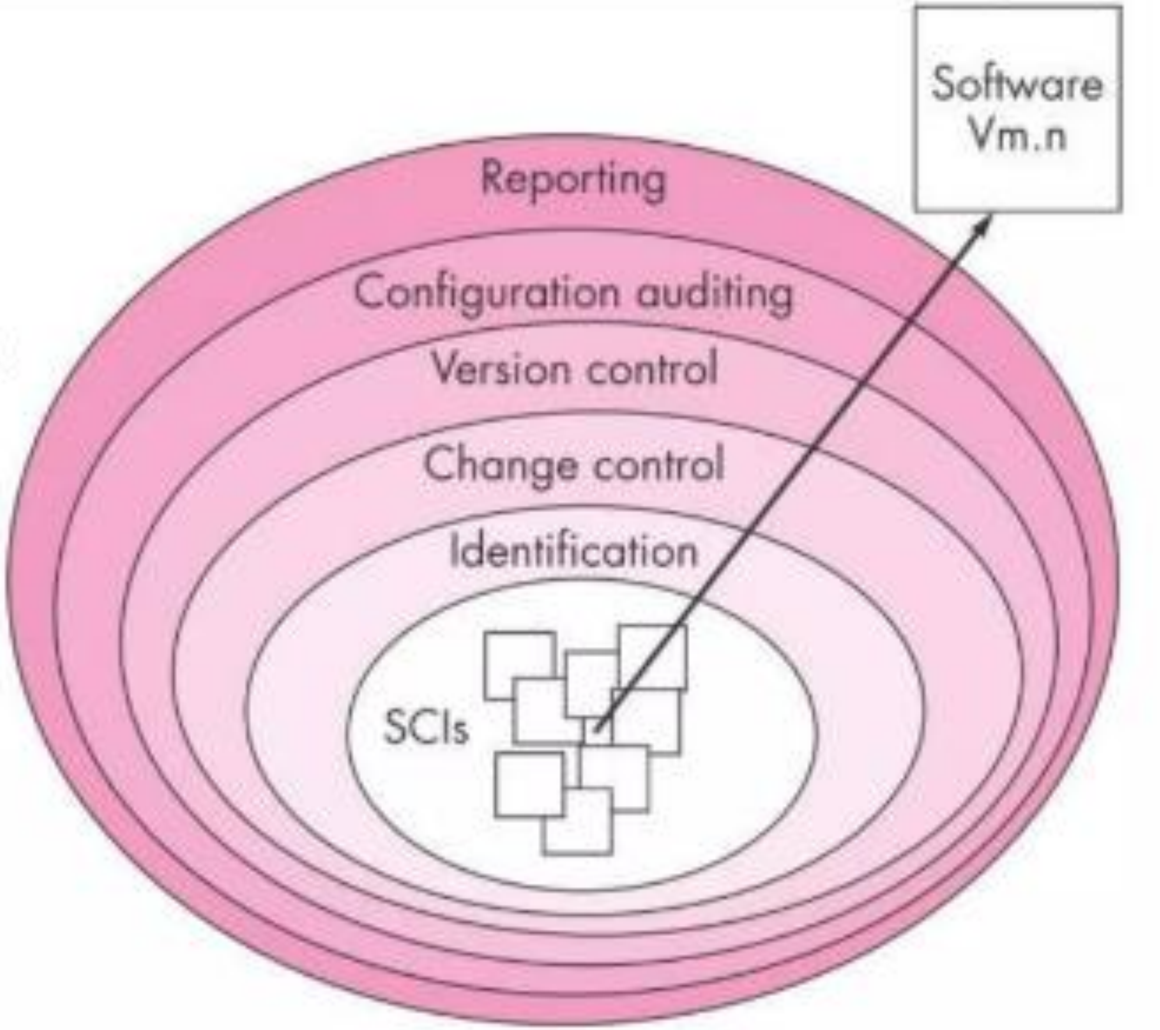
"A specification or product that ahs been formally reviewed and agreed upon, that thereafter serve as the basis for further development, and that can be changed only through formal change control procedure"

- Common Baselines :
  - Software Engineering /Specification
  - Requirement Analysis
  - Software design,
  - Coding,
  - Testing,
  - Release

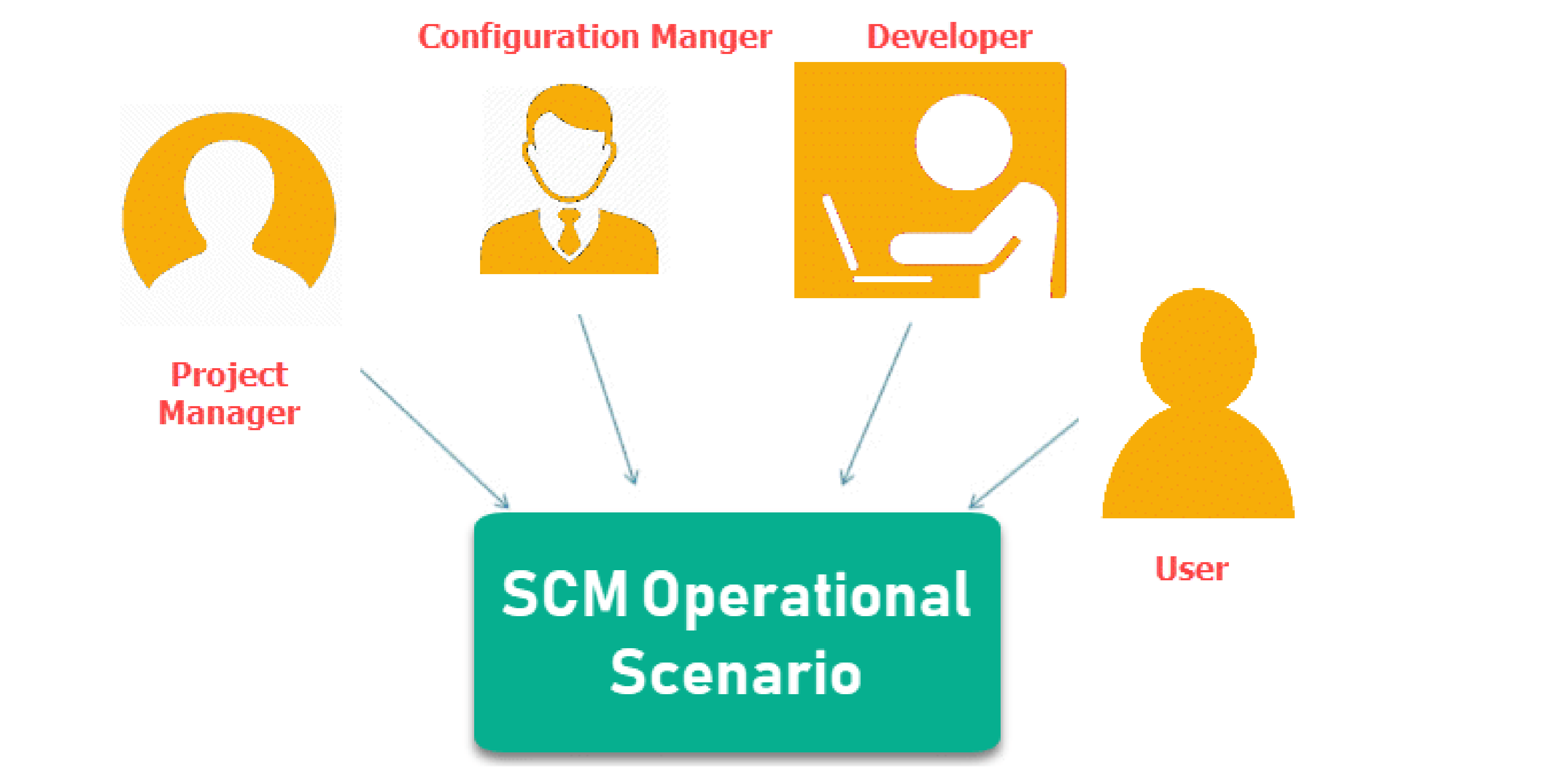
SCM Process

Primary Objectives:

1. To identify all items that collectively define the software configuration
2. To manage changes to one or more of these items
3. To facilitate the construction of different versions of an application
4. To ensure that software quality is maintained as the configuration evolves over time



Participants of SCM Repository



1. **Configuration Manager**
   - Configuration Manager is the head who is Responsible for identifying configuration items.
   - CM ensures team follows the SCM process
   - He/She needs to approve or reject change requests
2. **Developer**
   - The developer needs to change the code as per standard development activities or change requests. He is responsible for maintaining configuration of code.
   - The developer should check the changes and resolves conflicts
3. **Auditor**
   - The auditor is responsible for SCM audits and reviews.
   - Need to ensure the consistency and completeness of release.
4. **Project Manager:**
   - Ensure that the product is developed within a certain time frame
   - Monitors the progress of development and recognizes issues in the SCM process
   - Generate reports about the status of the software system
   - Make sure that processes and policies are followed for creating, changing, and testing
5. **User**
   - The end user should understand the key SCM terms to ensure he has the latest version of the software

SCM Repository

- The SCM repository is the set of mechanisms and data structures that allow a software team to manage change in an effective manner
- The repository performs or precipitates the following functions [For89]:
  - Data integrity
  - Information sharing
  - Tool integration
  - Data integration
  - Methodology enforcement
  - Document standardization

Functions of an SCM Repository

- Data integrity → Validates entries, ensures consistency, cascades modifications
- Information sharing→ Shares information among developers and tools, manages and controls multi-user access
- Tool integration → Establishes a data model that can be accessed by many software engineering tools, controls access to the data
- Data integration →Allows various SCM tasks to be performed on one or more CSCIs
- Methodology enforcement →Defines an entity-relationship model for the repository that implies a specific process model for software engineering
- Document standardization →Define objects in the repository to guarantee a standard approach for creation of software engineering documents

Change control

- Change control is a systematic approach that helps you manage incoming change requests throughout the life of your project.
- This approach focuses on thoroughly documenting and evaluating each change request to determine its feasibility and potential value. After this evaluation, the team can decide to approve, reject, or defer a project change until a later date.
- **Different Types of Change Requests**
  - Throughout the life of your project, there will be two kinds of changes that occur:
  - **Changes beyond your control**: weather events, team attrition, or employee sickness.
  - **Changes that are specifically requested**: Adding features to software applications, Updating network equipment, Installing system patches

Change control process

- **Step 1: Document the Change Request**
  - When a change request occurs, your first step is to categorize and record it. This is required even if you ultimately decide not to pursue it.
- **Step 2: Conduct a Formal Change Evaluation**
  - Your project team will meet and formally evaluate the change. Key questions to ask include:
  - Is the change justified?
  - What are the risks and benefits of making the change?
  - What are the risks of not making the change?
  - Do the risks outweigh the benefits?
  - If you decide to accept a change, you'll assign the next steps to a dedicated development team.
- **Step 3: Plan the Change**
  - The team that you task with developing the change will create a detailed plan outlining how it will be designed and implemented.
  - At this juncture, there should be methods in place to define and verify success.

Change control process

- **Step 4: Design Software Changes**
  - If the requested change affects your ERP software, the team will design the new program and test it for accuracy. If it passes successfully, they'll request approval and set a date for implementation.
- **Step 5: Conduct an Internal Software Review**
  - If the software change is approved, the team will proceed to implement the change. Then, they'll perform demonstrations for stakeholders so they can review and accept the final change.
- **Step 6: Conduct a Final Assessment**
  - If everything is implemented and reviewed successfully, the status of the change request will change from active to closed.

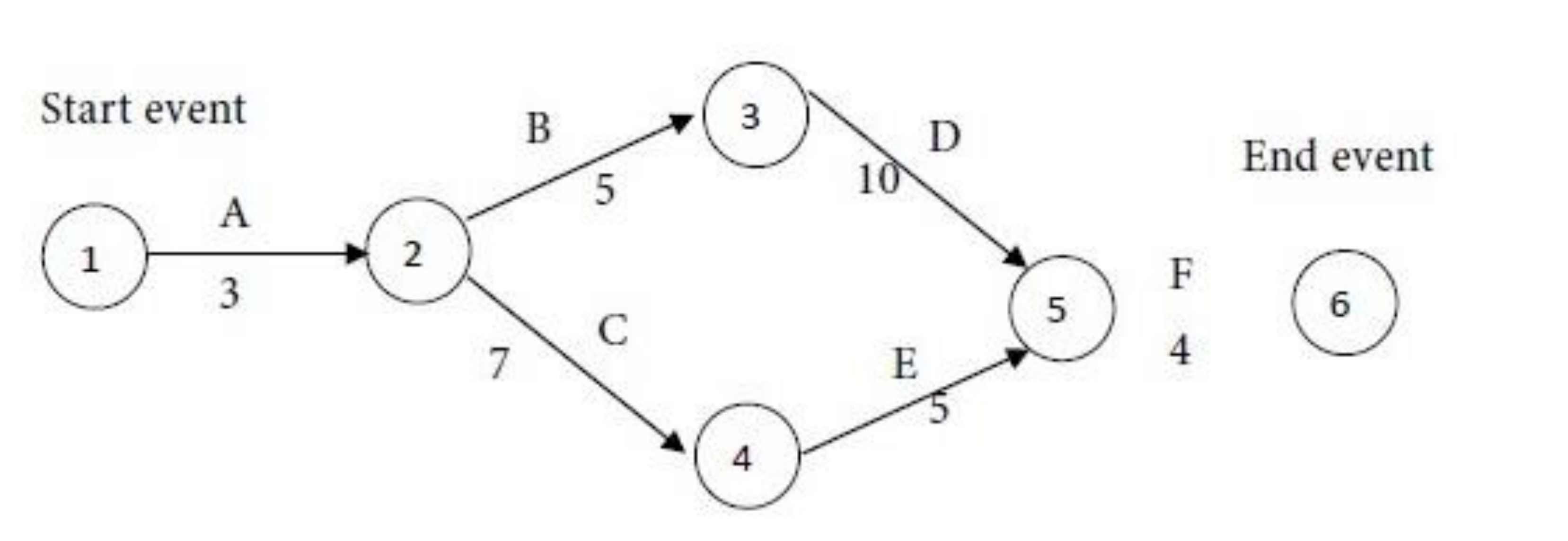
Chapter 2

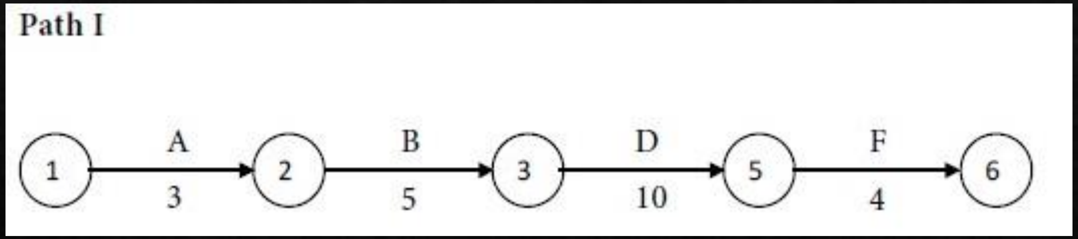
Critical Path : The critical path refers to the **longest stretch** of the activities, and a measure of them from start to finish.

The following details are available regarding a project:

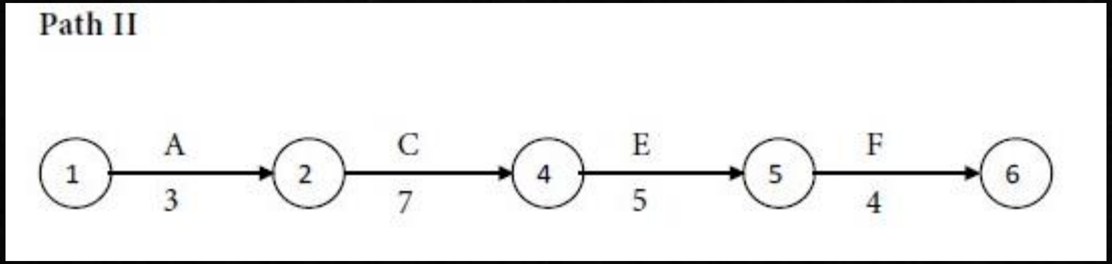
Activity	Predecessor Activity	Duration (Weeks)
A	-	3
B	A	5
C	A	7
D	B	10
E	C	5
F	D,E	4

Cont.





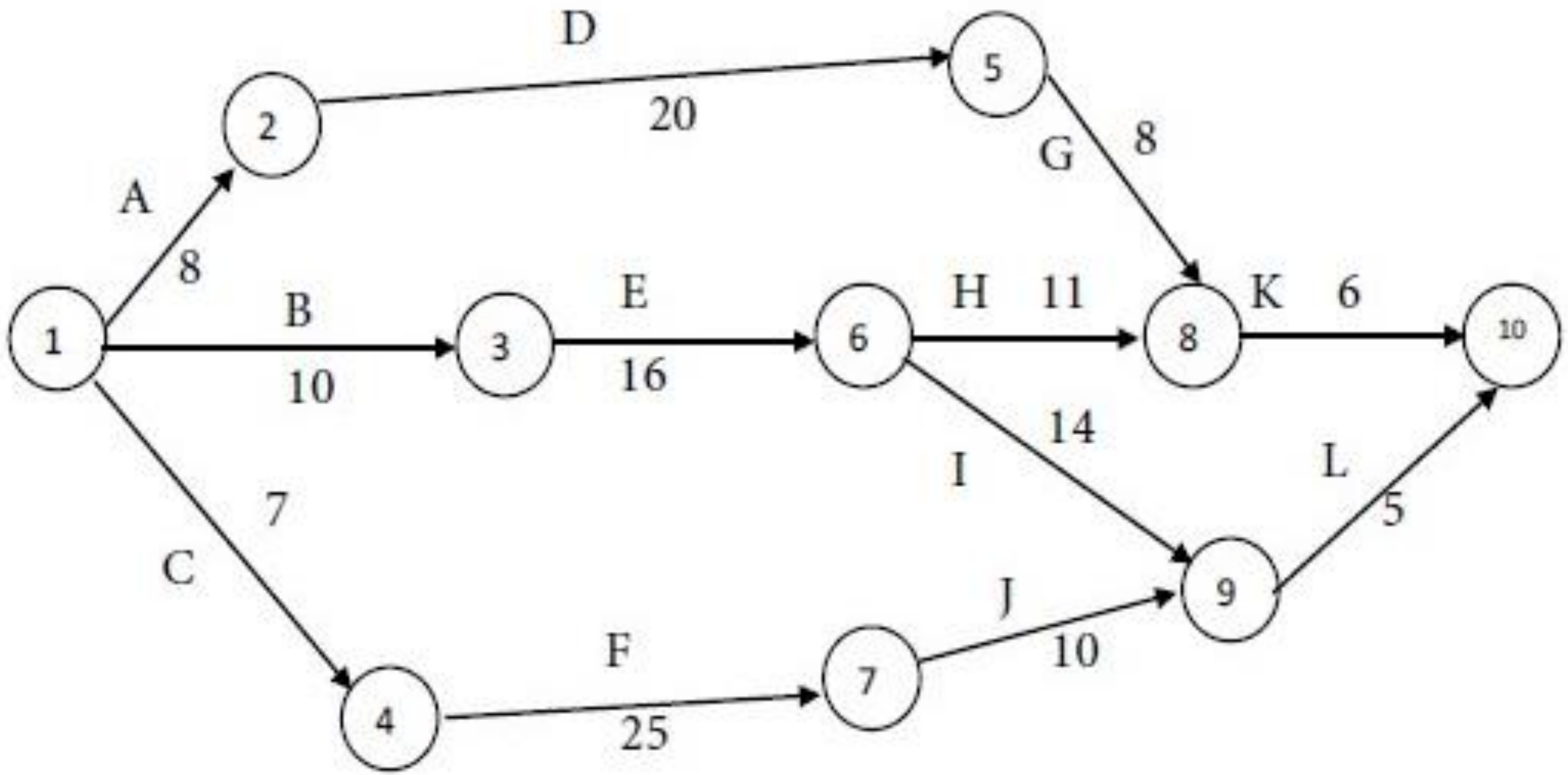
3+5+10+4=22 weeks



3+7+5+4=19 weeks

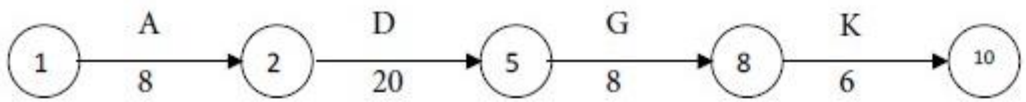
Hence critical Path is Path I with 22 Weeks.

Find out the completion time and the critical activities for the following project:



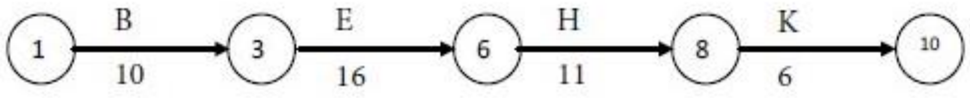
Solution

Path I



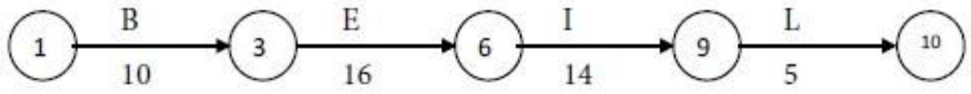
Time for the path = 8 + 20 + 8 + 6 = 42 units of time.

Path II



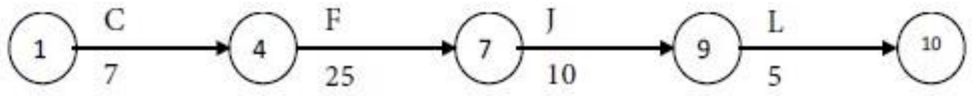
Time for the path = 10 + 16 + 11 + 6 = 43 units of time.

Path III



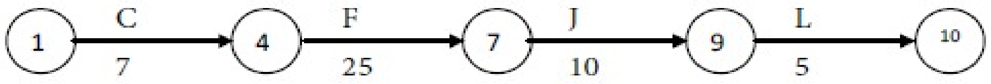
Time for the path = 10 + 16 + 14 + 5 = 45 units of time.

Path IV



Time for the path = 7 + 25 + 10 + 5 = 47 units of time.

Compare the times for the four paths. Maximum of {42, 43, 45, 47} = 47. We see that the following path has the maximum time and so it is the critical path:



The critical activities are C, F, J and L. The non-critical activities are A, B, D, E, G, H, I and K. The project completion time is 47 units of time.

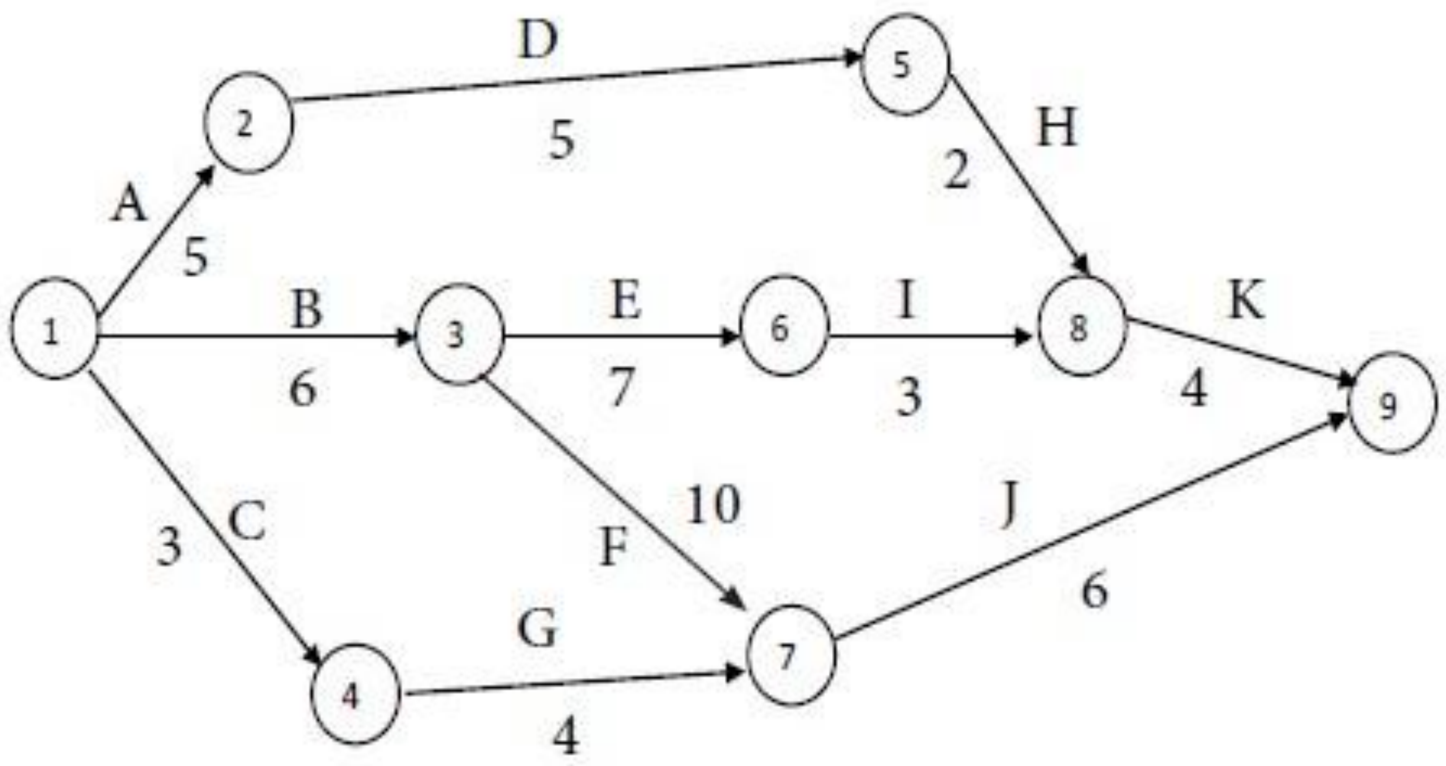
Cont.

Draw the network diagram and determine the critical path for the following project:

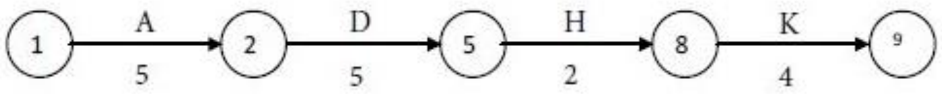
Activity	Time estimate (Weeks)
1- 2	5
1- 3	6
1- 4	3
2 -5	5
3 -6	7
3 -7	10
4 -7	4
5 -8	2
6 -8	5
7 -9	6
8 -9	4

Question and Solution

We have the following network diagram for the project:

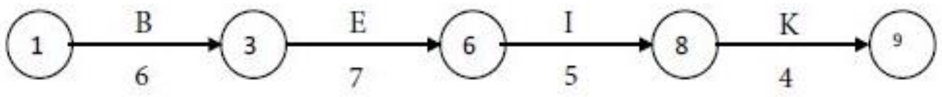


Path I



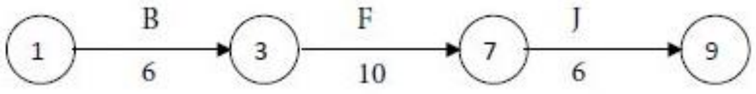
Time for the path = 5 + 5 + 2 + 4 = 16 weeks.

Path II



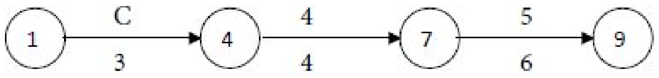
Time for the path = 6 + 7 + 5 + 4 = 22 weeks.

Path III



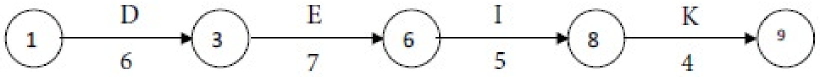
Time for the path = 6 + 10 + 6 = 16 weeks.

Path IV



Time for the path = 3 + 4 + 6 = 13 weeks.

Compare the times for the four paths. Maximum of {16, 22, 16, 13} = 22. We see that the following path has the maximum time and so it is the critical path:



The critical activities are B, E, I and K. The non-critical activities are A, C, D, F, G, H and J. The project completion time is 22 weeks.

PERT Networks (cont'd)

- Formula for Calculating Estimated PERT Times

$$T_e = \frac{T_o + 4T_m + T_p}{6}$$

T_e = Time estimate (average)

T_o = Time estimate (optimistic)

T_m = Time estimate (most likely)

T_p = Time estimate (pessimistic)

Standard Deviation of an Activity, $\sigma = \frac{P - O}{6}$

Variance of an Activity, $\sigma^2 = \left(\frac{P - O}{6}\right)^2$

Condition in Critical Path

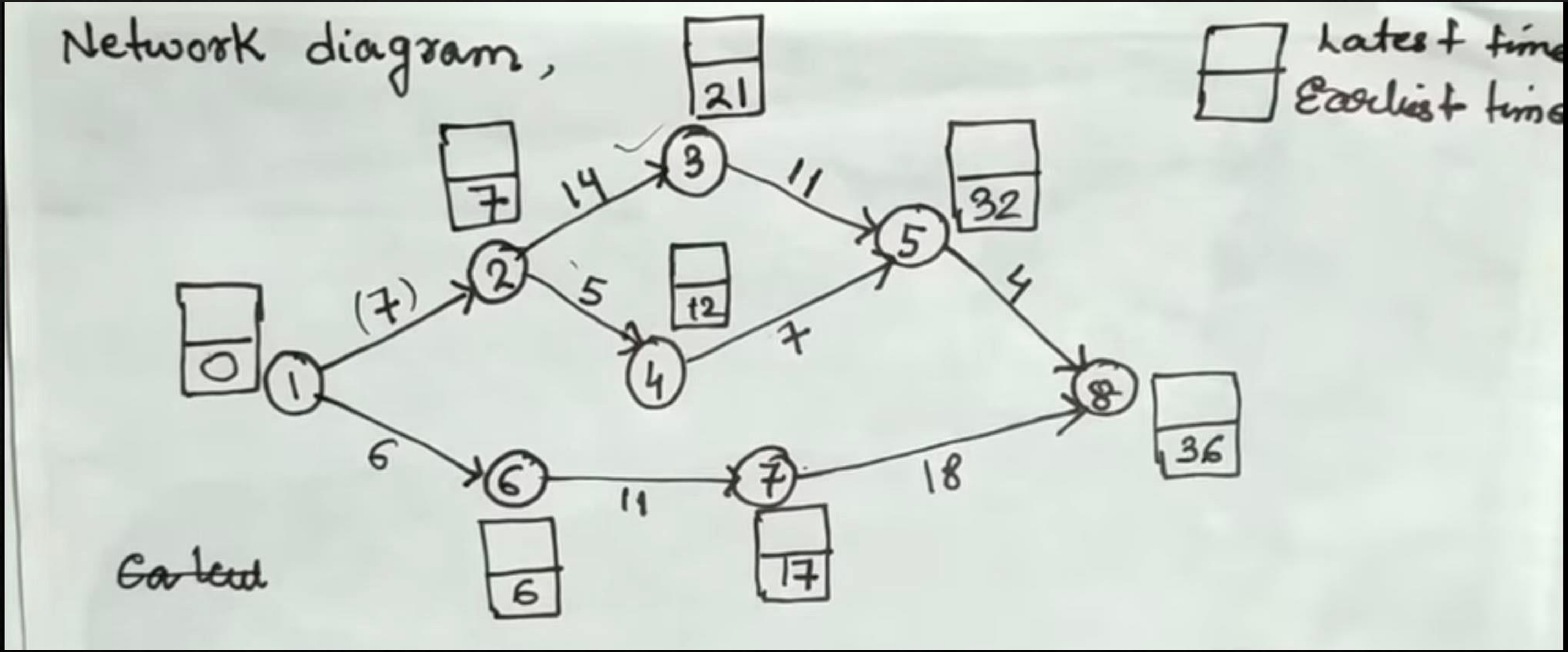
$$\left.\begin{array}{r} E_i = L_i \\ E_j = L_j \\ E_j - E_i = L_j - L_i = D_{ij} \end{array}\right\}$$

PERT (Program Evaluation and Review Technique)

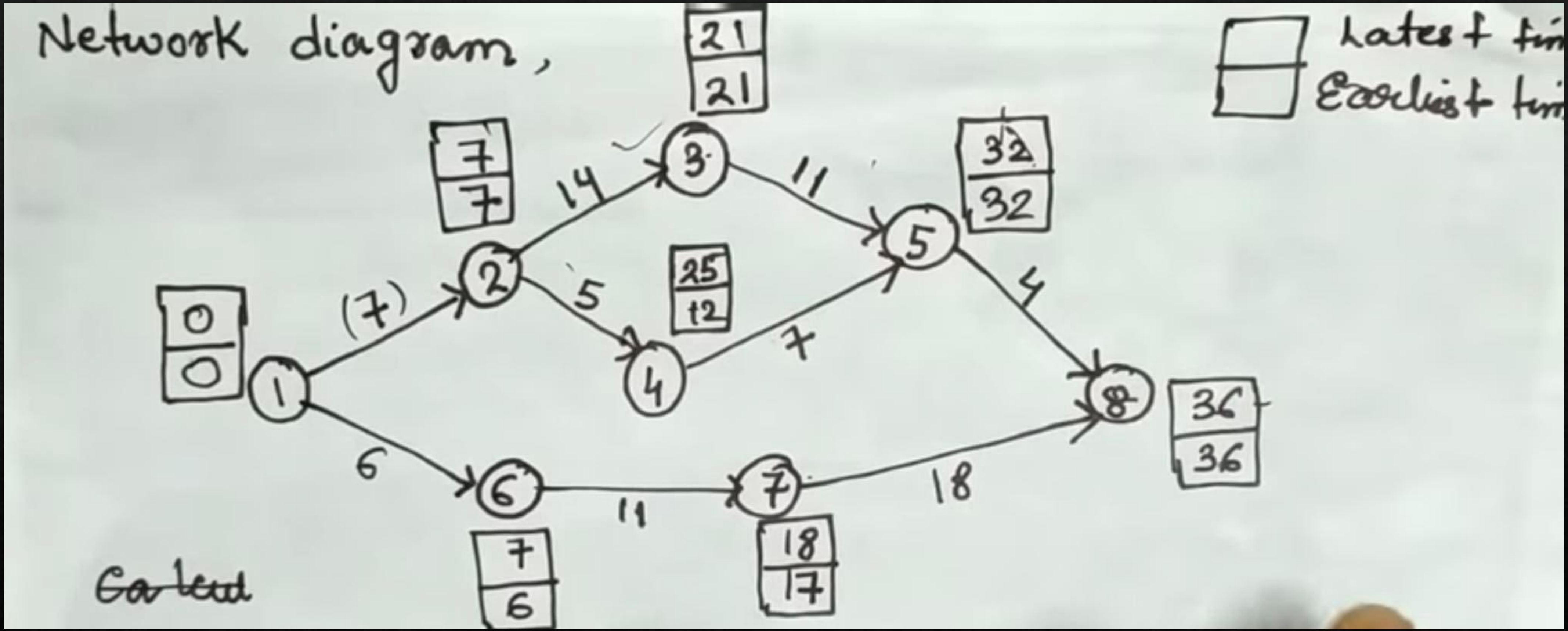
Activity	Estimated Duration (Weeks)		
	Optimistic time (To)	Most Likely Time (Tm)	Pessimistic time (Tp)
1-2	1	7	13
1-6	2	5	14
2-3	2	14	26
2-4	2	5	8
3-5	7	10	19
4-5	5	5	17
6-7	5	8	29
5-8	3	3	9
7-8	8	17	32

Activity	Estimated Duration (Weeks)			$T_e = \frac{T_0 + 4Tm + Tp}{6}$	$\sigma^2 = \left(\frac{T_P - T_0}{6}\right)^2$
	Optimistic time (To)	Most Likely Time (Tm)	Pessimistic time (Tp)		
1-2	1	7	13	7	4
1-6	2	5	14	6	4
2-3	2	14	26	14	16
2-4	2	5	8	5	1
3-5	7	10	19	11	4
4-5	5	5	17	7	4
6-7	5	8	29	11	16
5-8	3	3	9	8	1
7-8	8	17	32	18	16

Earliest Time



Latest Time



Network diagram,
Latest tim
Earliest tim
(7)
14
5
11
7
4
6
11
18
21
21
7
7
25
12
32
32
0
0
36
36
7
6
18
17

Critical Path=**1-2-3-5-8**

Expected project duration=7+14+11+4=36 weeks

Project length variance=4+16+4+1=25

Standard Deviation=5